The Portfolio – Work in Progress Continues

NoteMates: Your Study Buddy for Your Notes
Part 4 (Draft): Updating the App Build

Important Considerations

Three things to check before you submit:

- **Check that PartyRock can reach the internet.** I haven't confirmed which widgets or settings let an app pull in live web pages. If you don't see a web option when you build, the AWS documentation source may need to be text you paste in or upload yourself, and the box label ("Public pages online") should change to match.
- **Update the wording elsewhere.** Part 1 and Part 2 say answers come only from the student's notes, and the welcome message and chatbot prompt say "from your notes." If documentation is a source too, change those to "your notes or AWS documentation," and add a line in your Part 3 testing for a question answered from each source.
- **The responsible AI point still applies.** Web content can be wrong or out of date, so keep the instruction to say when an answer isn't in the sources, and compare a few answers against the current official AWS pages.

PartyRock App Flow

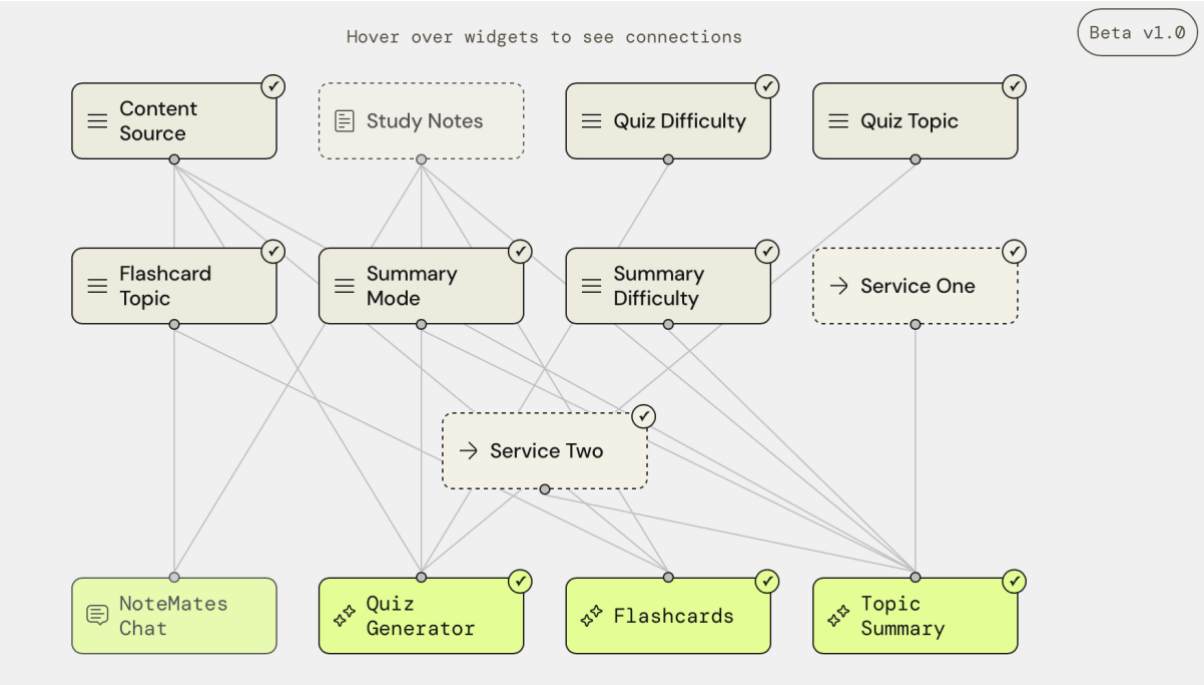


Figure: How NoteMates widgets are connected.

Notemates Flow Sources

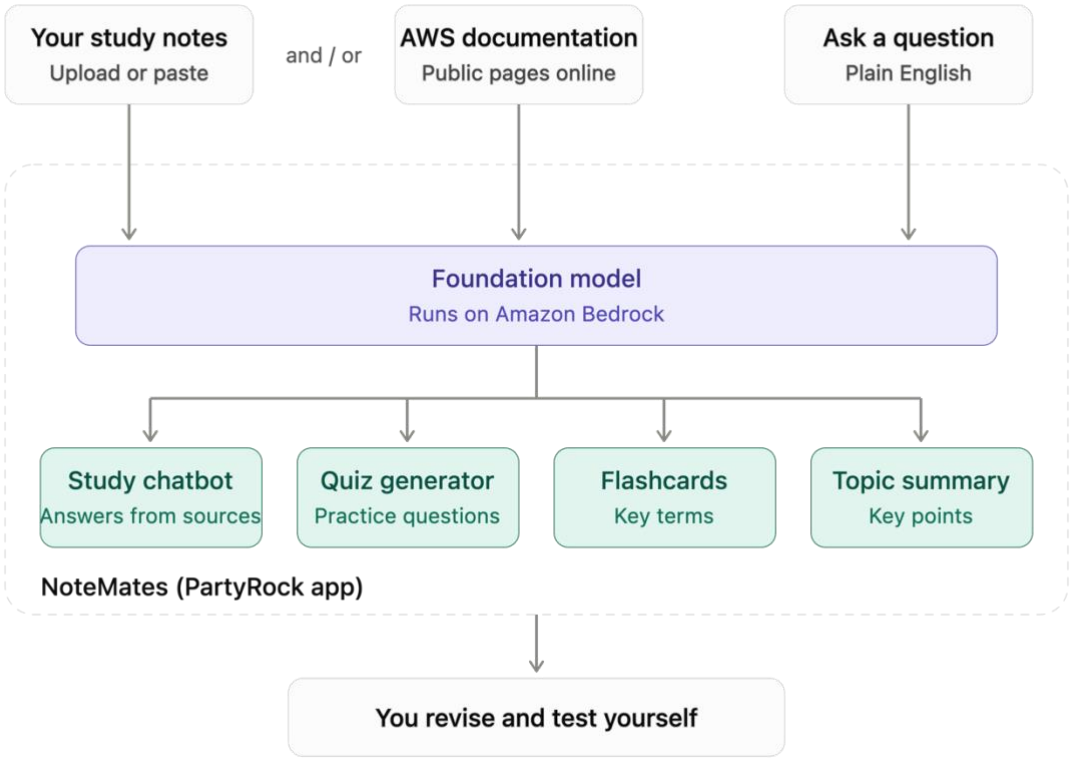


Figure 1: How NoteMates turns a student's study notes, AWS documentation, or both into answers, quizzes, flashcards and summaries.

PartyRock Flow Architecture

The *updated* welcome message

To make NoteMates feel like a study buddy, I re-wrote the welcome message:

"Hey there, welcome to NoteMates! 👏 Your notes and you make the perfect study pair.
I'm your study buddy for the AWS Cloud Practitioner exam. Here's what I can do:

- Answer questions straight from your own notes or AWS documentation
- Quiz you with exam-style practice questions
- Build flashcards for key terms and definitions
- Summarize a topic into quick review points

To get started: add your study notes, point me to AWS documentation, or use both, then try asking something like "What's the difference between S3 Standard and S3 Glacier?" or "Quiz me on storage services."

One promise: I answer from your notes or AWS documentation. If something isn't in them, I'll tell you instead of guessing, and it's always worth double-checking important facts on the official AWS website.

Ready when you are. What are we studying today?"

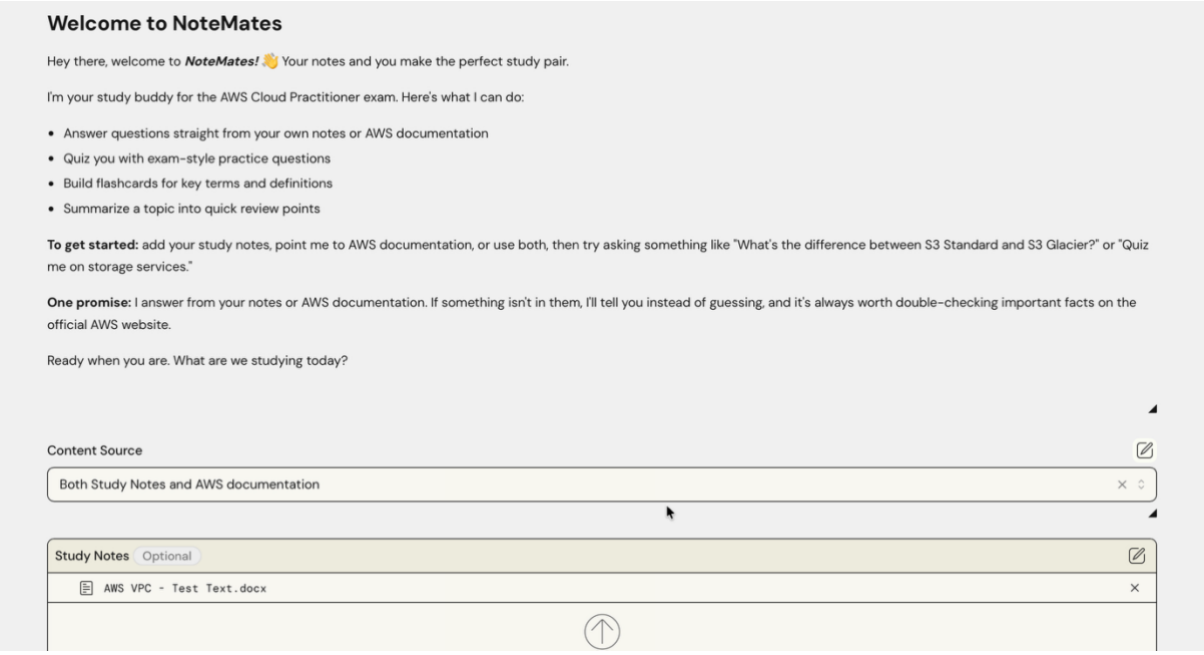


Figure 2: The NoteMates app in Amazon PartyRock.

Content Source Section

Prompt: *“Can you provide me with a bullet point summary of the information I uploaded in the study notes section?”*

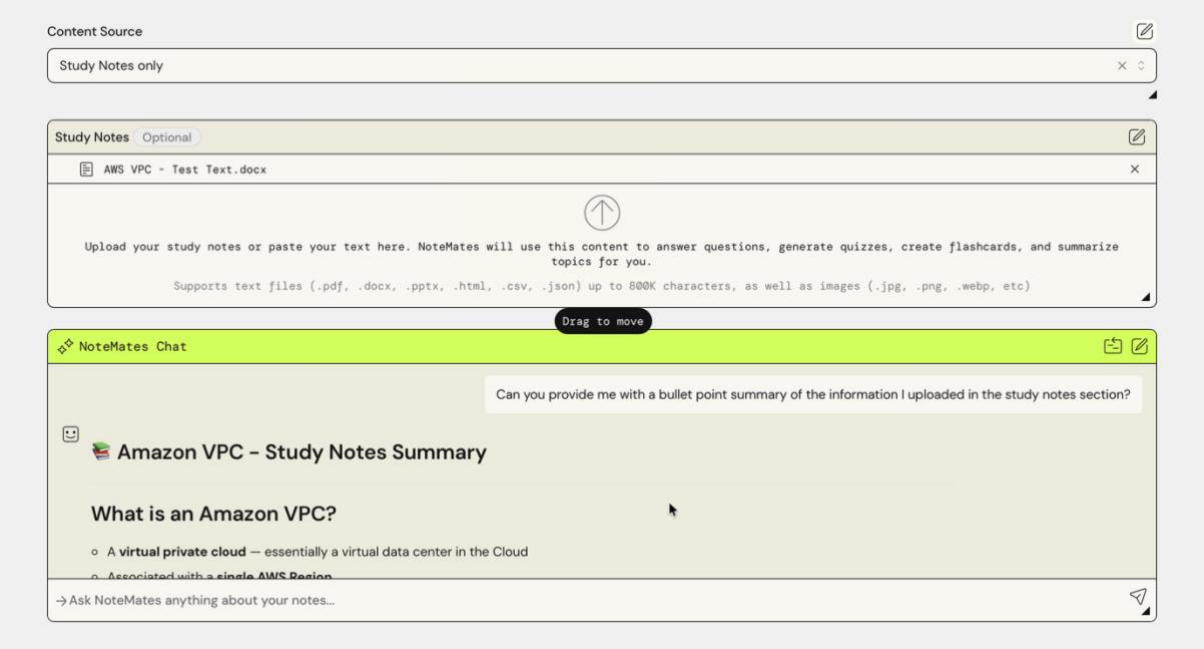


*Figure 3: A content source section – **AWS VPC – Test Text.docx***

Generated Quiz, Flashcard and Topic Summary

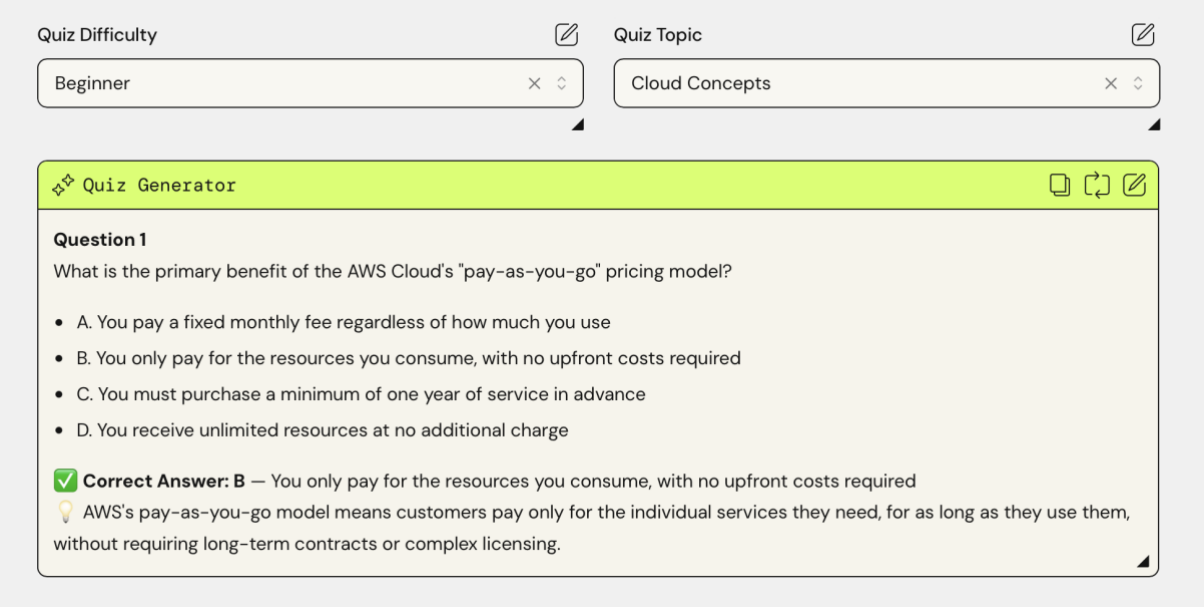


Figure 4: A generated quiz section

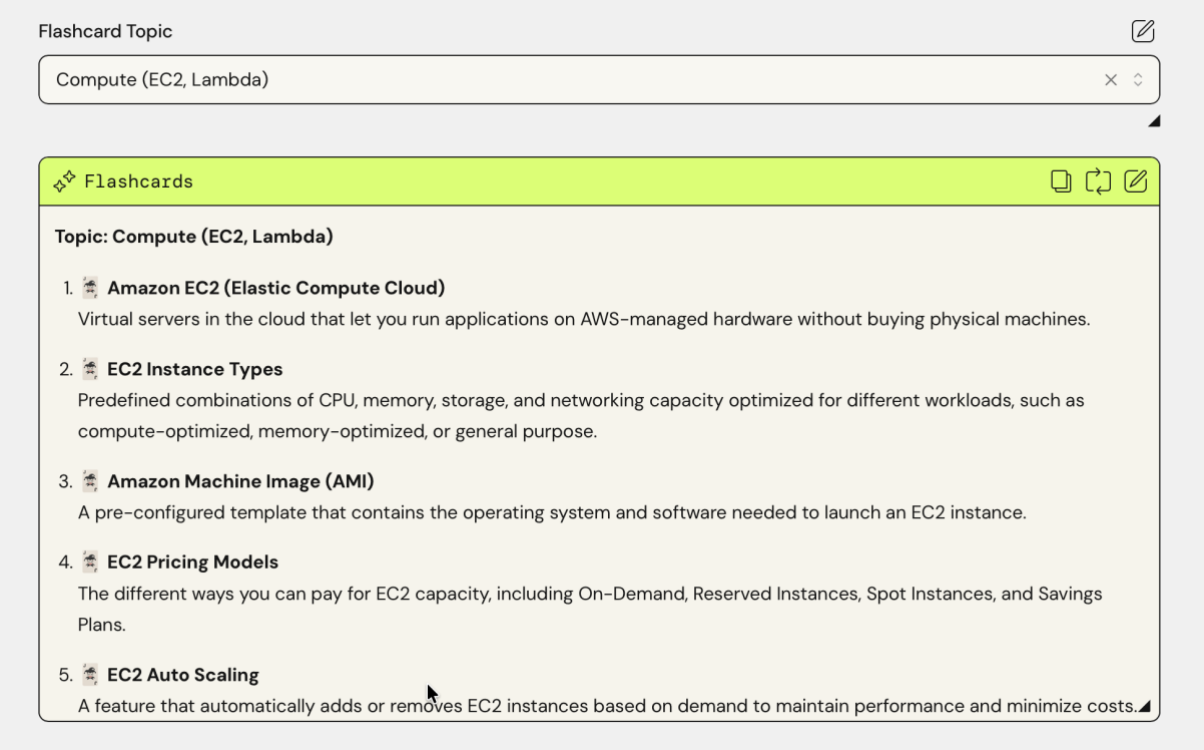


Figure 5: The flashcards section

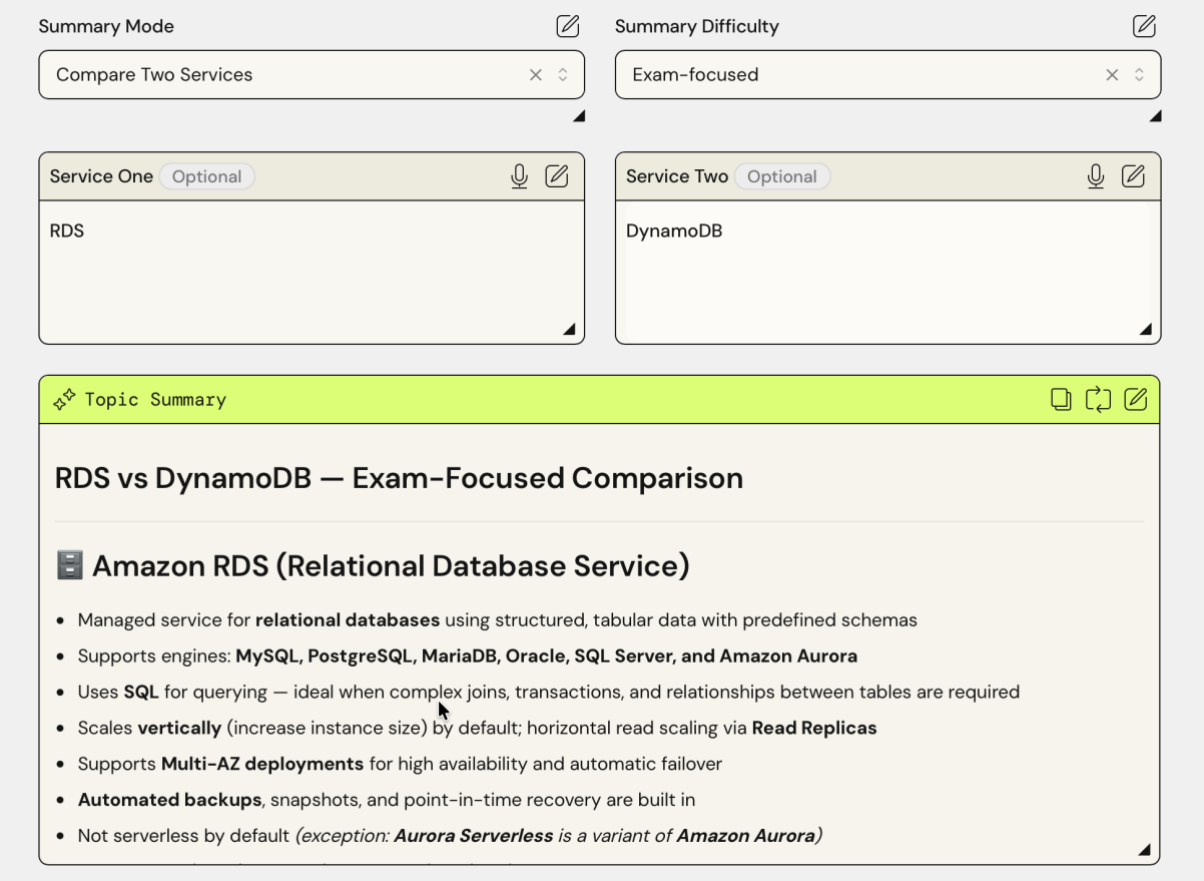


Figure 6: A topic summary section

Prompts used for improvements

- ***Updating Correcting AI Widgets stalling issue across sections:***
  "The app should not wait for @Study Notes to get started in the other widgets. If nothing is uploaded in the @Study Notes widget then AWS Documentation should automatically be used"

- ***Updating difficulty across sections:***
  "The difficulty mode selection options are Foundational, Associate, and Professional which is more working-professional focused. This needs to be replaced by Beginner, Intermediate, and Exam-focused across all relevant widgets with a difficulty dropdown because this is aimed at students working towards their professional careers."

- ***Increasing the number of Quiz questions:***
  "The Generated Quiz is only 5 multiple choice questions and I would like this to be 10."

Testing: grounded vs. direct answers

I tested the NoteMates Chatbot in two ways: questions answered from my uploaded notes, and the same questions asked of the model directly with no notes.

Test 1: Question from my notes (Study Notes only)

- ***Question:*** Does Amazon VPC span across multiple regions?
- Did it match my notes? ***Yes***
- ***NoteMates' answer:***

  🌐 **Amazon VPC & Regions**

  Based on your study notes — No! ❌

  Your study notes clearly state:

  > *"An Amazon VPC lives in **one Region**"*

  Key Points to Remember

- A VPC is **locked to a single AWS Region**
- However, within that Region, your VPC **can span multiple Availability Zones**
- This is actually how AWS recommends you build for **resiliency and availability**

  Quick Breakdown

Resource	Scope
VPC	**One Region only**
Subnet	**One Availability Zone only**
Route Table	Can associate with **multiple subnets**

  💡 **Exam Tip**

  This is a **commonly tested concept** on the AWS Cloud Practitioner exam!

Remember:

- **VPC = Regional** 🗺️
- **Subnet = Single AZ** 📍
- If you need resources across Regions, you would need to create a **separate VPC in each Region**

⚠️ Don't confuse **multi-AZ** (within one Region) with **multi-Region** — they are very different concepts!

Test 2: Same question, no notes (AWS Documentation only)

Answer:

🌐 **Amazon VPC & Regions**

Short Answer: No! ❌

An Amazon VPC does NOT span across multiple regions. Each Amazon VPC is associated with a single AWS Region only.

Key Points to Remember

Concept	Scope
Amazon VPC	**Single Region**
Subnet	**Single Availability Zone**
Default VPC	**One per Region**

How It Works 🏗️

- When you create an AWS account, separate default VPCs are automatically created in each supported Region.
- If you need resources in multiple regions, you would need to create a separate VPC in each Region
- For added resiliency, a VPC can be divided into multiple subnets across different Availability Zones within that single Region

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💡 Exam Tip

A common trick question on the AWS Cloud Practitioner exam is confusing **VPC scope (Regional)** vs **Subnet scope (single AZ)**. Remember:

- **VPC = One Region**
- **Subnet = One Availability Zone**

Difference from Test 1:

[e.g., more general, different wording, included things not in my notes]

Test 3: a question my notes don't cover (Study Notes & AWS Documentation)

- ***Question:*** *"Tell me more about AWS IAM?"*
- ***What happened:*** The chatbot provided a comprehensive answer from AWS Documentation because the question I asked did not reference information I had in my notes.

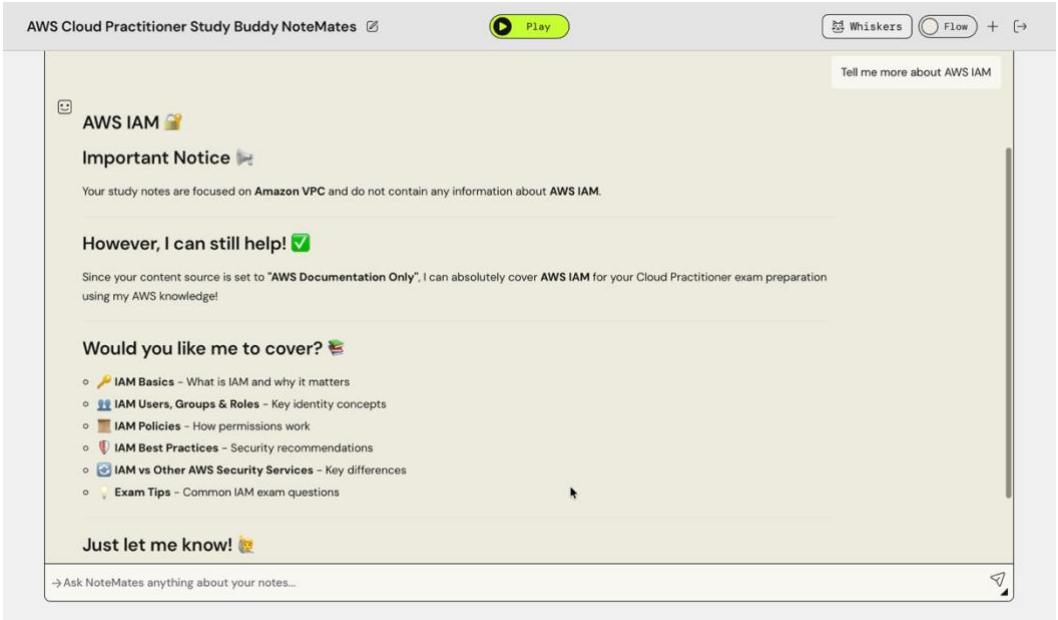


Figure 7: A Chatbot answer screenshot

New Issues Identified

1. The widgets stall when notes are empty and the source is "Study Notes only". I need to update all AI widgets to automatically fall back to AWS documentation when no notes are provided.
2. Quiz Generator, Flashcards, and Topic Summary widgets all have updated prompts with explicit fallback logic — if "Study Notes only" is selected but nothing is uploaded, they automatically use AWS documentation instead of stalling
3. The @Quiz Generated widget outputs a long list of text. Ideally, it would let users select an answer, show only one question at a time, and move on to the next question only after a selection is made. However, this isn't possible in PartyRock. AI Output widgets display all their generated text at once and can't respond to button clicks or answer selections, so a user's choice can't trigger the next output.
4. The upload section for the User Study Notes does not work, the only way to upload notes is by drag-and-drop the file into the upload field.

Key Changes

1. The study notes upload has been optional, so that official AWS Documentation is referenced if no notes are uploaded.
2. The welcome/introduction section is more robust and welcoming and includes a subtle “how to” for the app.
3. Quiz Difficulty and Summary Difficulty dropdowns now offer Beginner, Intermediate, and Exam-focused (defaulting to Beginner). Both ***Quiz Generator*** and ***Topic Summary*** prompts now interpret these levels meaningfully:
    a. Beginner → simple language, core definitions, no jargon
    b. Intermediate → scenario-based, focuses on how things connect
    c. Exam-focused → mirrors real exam phrasing, includes plausible distractors and flags common traps
4. The Generated Quiz is only 5 multiple choice questions and I have changed this to be 10.